

From theory to application: can we build a realistic model of internal migration?

Overview

In 1876, Ernst Ravenstein set out his “laws of migration” (Rees and Lomax 2019), which have informed an entire field of migration research. This paper borrows from that field, taking relevant theory and analysis and turning it in to a model of internal migration in the UK. The model will be assessed and validated against observed migration data to assess how well the theory has been translated into the model. If it can be validated the model will be integrated into a dynamic microsimulation pipeline.

Note the below is specified for **individuals**. A household specification can be developed alongside

Model structure and workflow

Step 1 – Baseline population	Step 2 – propensity to migrate	Step 3 – assign moving individuals to model	Step 4a – Local model	Step 4b – National model	Step 4c – National model allocation	Step 5 - Validate	Step 6 – refine and iterate
Initiates a population of individuals in LSOAs	Assigns each individual with a propensity to migrate	Two models - local or national model are specified	The local model is a gravity model of LSOA-LSOA	The national model is propensity/rules based of LSOA - LAD	Allocates inter LAD moves to LSOAs	Validate against observed census migration data	Improve model fit and repeat
Possible model(s): - uses COMPASS population Or uses census data and IPF to create individuals	Possible model(s): Probability distribution based on migrant age profile	Possible model(s): Deterministic – based on some sensible cut-off	Possible model(s): A spatial interaction or gravity model	Possible model(s): Probability distribution across an origin-destination matrix	Possible model(s): Simple rules based on pop. Density Schelling Cellular Automata	Possible model(s): Correlation/TAE – modelled vs observed	Possible model(s): : N/A

Step 1 – baseline population

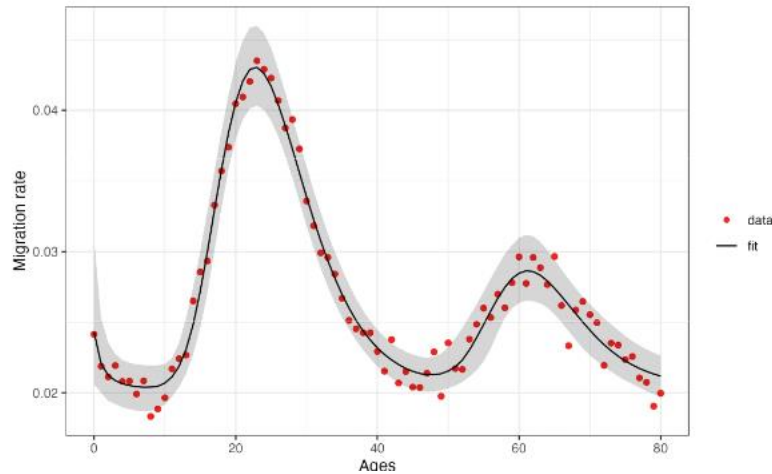
Imports population from COMPASS. Minimum requirements are LSOA area of residence, age and sex. For improvements, further attributes such as education and employment are needed. Draws on the theory spelt out in our work (Hoehn et al. 2024) on the rationale and validity of creating individual level data (beyond the counts available in e.g. the Census).

Step 2 – propensity to migrate

A propensity to migrate in the current time period is assigned to each individual in the baseline population. This assignment adapts the model migration schedule originally proposed by Rogers and Castro (1978) and subsequently updated by Wilson (2010). A

rather nice Bayesian implementation of the Roger Castro model has been implemented by Yeung et al. (2023):

Figure 5: Example of result from Bayesian estimation of Rogers–Castro model parameters



Source: Yeung et al. (2023)

The alternative is simpler: we can borrow the observed data from UK Census to produce the probability distribution. e.g. see figure below that I produced for my PhD (suggest fig. b is most complete, <https://etheses.whiterose.ac.uk/id/eprint/5839/>):

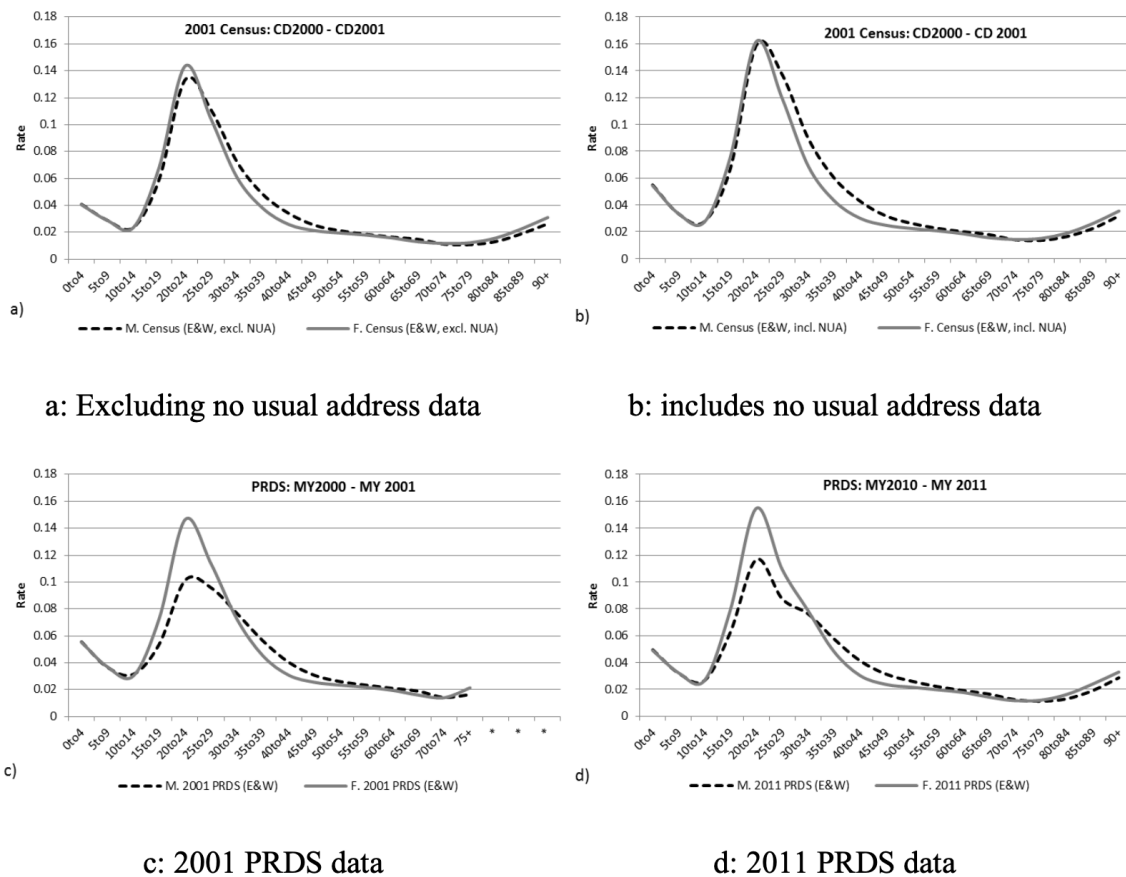


Figure 7.4: A comparison of migration schedules in England and Wales from the 2001 Census and PRDS data for 2000/01 and 2010/11

However, having some implementation of the Rogers Castro model might be preferable as it would not require input data.

Step 3 – assign moving individuals to model

Because the process and motivation for a local (short distance) vs a longer distance move varies we should allocate individuals to one of the two models. Short distance moves are very sensitive to distance, driven by housing, family and area quality motivations. Longer distance moves are less sensitive to actual distance and driven by employment and education motives. See below from Thomas (2019) and Thomas et al. (2019).

[add here Hurdle Model and International Migration competing risks]

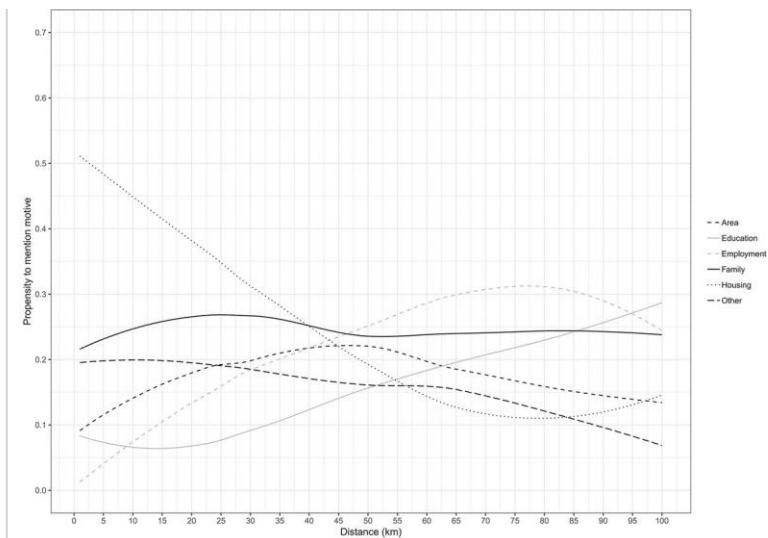
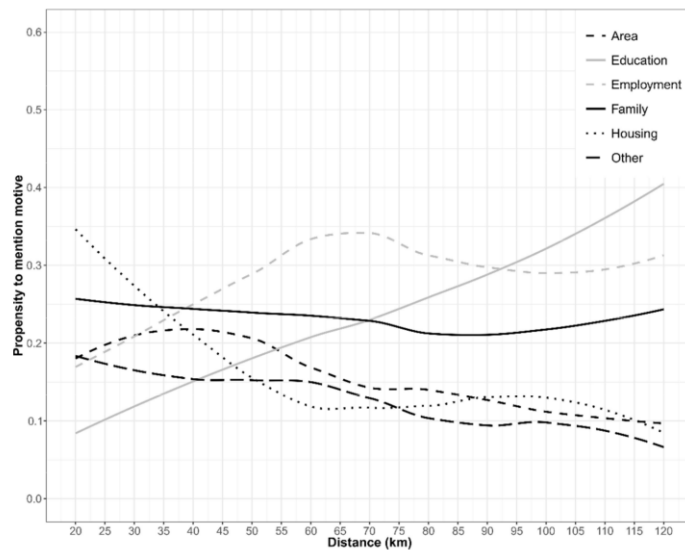


Figure 1

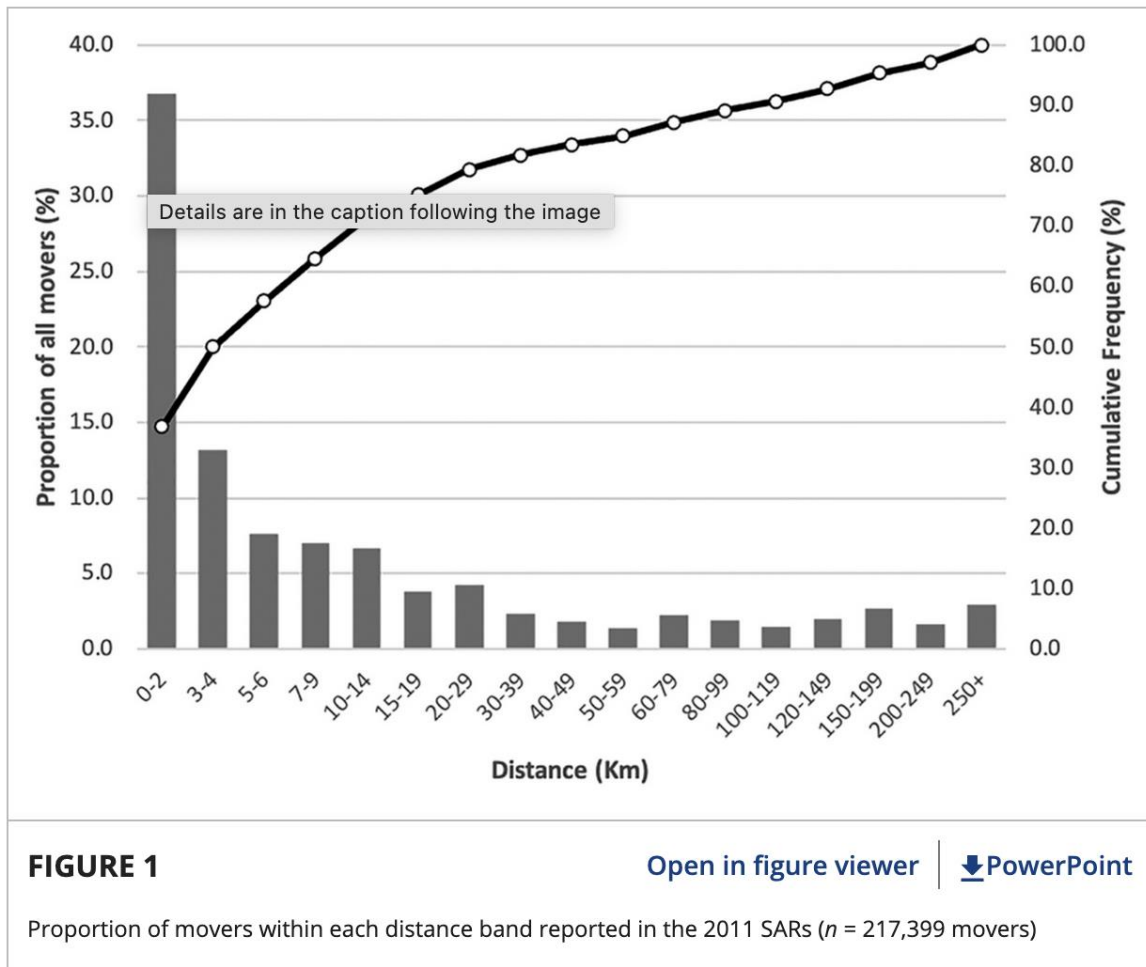
[Open in figure viewer](#) | [PowerPoint](#)

Variations in stated migration motives by distance (pooled UKHLS 2009–2016). Source: University of Essex. Institute for Social and Economic Research, NatCen Social Research, Kantar Public. (2018). Understanding Society: Waves 1–7, 2009–2016 and Harmonised BHPS: Waves 1–18, 1991–2009: Special Licence Access, Census 2001 Lower Layer Super Output Areas. [data collection]. 9th Edition. UK Data Service. SN: 6670, <http://doi.org/10.5255/UKDA-SN-6670-9>. Estimates account for complex survey design and non-response using the “survey” package in R (Lumley, 2018). N.B. $n = 15,083$, based on a pooled cross-sectional sample of UKHLS wave pairs covering the period 2009–2016, includes individuals (aged 16+) who provide a motive for moving at t_1 and move any distance between waves t_0 and t_1

Figure 1: The propensity to mention a motive for moving within the United Kingdom



One way of doing this initially is a random allocation based on a distance cut-off. We know that 50 per cent of moves are under 4km, and 75 per cent of moves are under 29km (Lomax et al. 2021) – so these might be a useful starting point (see figure below).



We might also want to stratify this by age and sex as we know the odds of moving over different distances varies by demographic attributes (also reported in Lomax et al 2021). Perhaps hold for a future iteration. Movers in the local model (short distance) are treated differently to movers in the national model (long distance).

Step 4a – Local model

Because distance, housing, area and family are important for local moves, we can specify a gravity model for these movers. Originally specified in a migration context by Zipf (1946) in a study of migration between US cities they have been extended to a wide range of migration contexts. Gravity models and Spatial Interaction Models (SIMs) are used interchangeably in the migration literature (where the SIM is the statistical form of the Gravity Model) and at their most basic include population size, distance and some measure of competition or attractiveness. They can also integrate other covariates as appropriate. For a couple of examples see Dennett and Wilson (2013) and Congdon (2010). This model can be applied directly at the LSOA level, where the destination options are all LSOAs but with the highest probabilities in the nearest LSOAs.

Step 4b – National model

In the national model, distance plays a less important role as education and employment become dominant. At longer distances, human capital (Sjaastad 1962) becomes the dominant narrative in the literature. People move to accumulate knowledge and skills which enhance their capital, hence the motivations of education and employment. This model could be far more rules based, for example the availability of jobs, education establishments as well as other factors TBC at destination. Distance is less important because once someone makes the decision to move there are similar opportunities in multiple urban centres. This model will be more difficult to align with the literature than the local model but should be straightforward to parameterise if we e.g. think of it as a set of weighted OD matrices for each age and sex group.

Origin/Dest	1	2	3	4	5	6
1	X	0.09	0.12	0.02	0.21	0.3
2		X		0.05	0.002	0.06
3			X			
4				X		
5					X	
6						X

The matrix would look different for young people about to enter education or first job vs retirees, where the former is heavily weighted towards big cities and the latter to amenity rich locations.

Destinations here are local authority districts. The LSOA allocation comes next.

Step 4c – National Model allocation

Because those migrants who enter the national model (step 4b) end up in a destination local authority district, they need one final allocation to an LSOA within that district. This allocation could be rules based (e.g. based on population density of LSOAs, i.e. replicating existing distributions) or more complex model. For example a Schelling model (Hatna and Beneson 2014) based on individual migrant attributes and the attributes of the possible destination LSOAs, or a cellular automata (CA) land use approach. Such CA approaches have been used in New Zealand (Cameron and Cochrane 2017) and the UK (Lomax et al. 2022).

Step 5 - Validate

If steps 1-4 have been implemented on a baseline population using rules derived from the theoretical literature there is data we can use to validate against. The Census 2011 and 2021 contain origin-destination matrices of migration at LAD level (by age and sex) and at LSOA level (aggregate). The question to answer is: does our translation of theory into model replicate the observed data, and how good is the fit? If it is good then we can integrate into our dynamic model if not it will need to be refined.

Step 6 – refine and iterate

This step involves estimating based on observed data, refining our approach and iterating through the model steps.

Ideas for followup

- Social Frontiers (Diasporas)
- Household 'musical chairs' model
- International model integration within 'hurdle model' framework

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